

Josh Wang

Software Developer

Specializing in React / Next.js architecture, AI-assisted development workflows, and high-efficiency product delivery

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Summary

10+ years of experience building React / Next.js applications across SaaS, cybersecurity, internal platforms, and mobile products. Led frontend architecture and delivery for 7+ web systems and 2 mobile apps using TypeScript, CI/CD, monorepo, and AWS. Focused on scalable frontend systems, AI-assisted development workflows, and legacy system modernization.

Key Achievements

- Reduced cloud infrastructure costs by 90%
- Supported internal systems across 50+ countries
- Built 13-language website architecture
- Reduced AI-assisted POC validation cycles from weeks to 1.5 days
- Consolidated CI/CD branches from 5 to 2
- Patched critical CVE-2025-29927 vulnerability

Experience

Frontend Architect (Contract-based)

Jun 2023 – Present

Independent

CI/CD Architecture & Security | Data Analytics SaaS Startup

- Restructured CI/CD pipeline from 5 fragmented branches to 2, optimizing monorepo structure and reducing operational overhead
- Planned and integrated automated testing architecture with Claude + Playwright MCP for Gherkin script generation, improving QA acceptance efficiency
- Performed critical security patching under time pressure, resolving CVE-2025-29927 and strengthening system security posture

Map System Development & Technical Debt Cleanup | Major Real Estate Group

- Served as technical lead to fill development gaps, delivering complex map POI filtering features on tight deadlines
- Adopted Claude Code AI-assisted development, reducing POC validation from weeks to 1.5 days
- Led systematic module decomposition and refactoring to resolve legacy architectural debt at the root

AI-Assisted Engineering Workflow & MCP

- Built reusable AI-assisted development workflows with Claude Code and MCP, accelerating project iteration cycles
- Explored Model Context Protocol (MCP) applications to improve AI collaboration accuracy across large codebases
- Established automated testing and acceptance pipelines (Playwright, Jest, Cucumber), reducing regression risk and improving release stability

Real-time Trading Platform | Vue 3 + Nuxt.js

- Built a real-time trading platform with Vue 3 / Nuxt 3 / Pinia
- Integrated TradingView Widget and ECharts for real-time price feeds, order book visualization, and data analytics

Front-End Developer (Platform & Architecture)

Oct 2021 – Jun 2023

TeamT5 (Cybersecurity)

ThreatSonar Product — Frontend Architecture Planning

- Led frontend architecture planning for frontend/backend separation, including page structure and data flow migration strategy
- Produced frontend technical specifications and architecture design documents as the foundation for implementation

Official Website

- Led cloud architecture migration from Firebase/Cloudinary to AWS S3/CloudFront + Next.js, reducing cloud costs by 90% and improving reliability
- Refactored 50%+ legacy codebase, improving maintainability and reducing resource overhead
- Implemented file-upload system with AWS SDK / TypeScript / Pica.js, providing auto-thumbnail and cloud upload functionality
- Conducted early ChatGPT-powered SEO crawler testing; findings presented to and recognized by executive team

CYBERSEC 2022/2023 & InterOp23 Event Pages

- Built event pages with Preact.js + Tailwind CSS / CSS Module, deployed via AWS S3 + CloudFront
- Identified plaintext password transmission and auth design flaws in external vendor systems; report escalated to leadership and drove remediation

Front-End Lead

May 2019 – Oct 2021

OEK Group

B2B/B2C Back-office System

- Led frontend rebuild using React Hooks and Context API, covering internal systems across 50+ countries
- Designed 100-field single-page forms using Formik / Yup / React.memo with strict validation and performance control
- Implemented real-time messaging and notifications via WebSocket
- Proposed and adopted GitHub Actions CI/CD pipeline as an early adopter upon its launch in Aug 2019

Website Redesign / Internal Training

- Took over external vendor's HTML/CSS and completed Next.js + i18n architecture (13 languages) within one month
- Led Pull Request Code Review culture to improve code quality and ensure requirement alignment

Web Developer

Jul 2014 – Apr 2019

Startups (Fullinn, CatFi, myTreat, Ecowork)

Frontend Architecture & SEO | Fullinn

- Led frontend architecture using Next.js SSR to build an SEO-optimized website builder for hotel operators
- First client ranked on Google's first page within one week of launch
- Introduced Flow.js and ESLint to reduce Code Review overhead and improve code reliability

Mobile & Cross-platform Development | Fullinn, CatFi

- Built cross-platform mobile apps with React Native, Redux, and Redux-saga for state and side effect management
- Integrated Firebase Push Notification and Code Push to reduce deploy costs and streamline push services
- Adopted Fabric and OneSignal for crash reporting and push; integrated WebRTC / RTMP streaming

Full-stack Development & API Design | Fullinn, myTreat, Ecowork

- Built OTA platform and order system with React.js / GraphQL; migrated project from Redux to Apollo 2.0
- Led initial REST API format design, establishing API conventions to streamline backend development
- Delivered full-stack features and standalone customer service back-office using Rails and Angular.js

Testing & Dev Workflow | Ecowork

- Implemented Cucumber and RSpec to fulfill client test report requirements; assisted DevOps in building automated test environments
- Introduced Vagrant-based deployment workflow to unify dev environments for faster onboarding

Projects

PickleScout — LLM-Driven E2E Test Generation Agent

May 2026

Open Source Side Project | github.com/iskWang/PickleScout

- Completed full POC in one week using AI-assisted workflow; open-sourced with browser agent, LLM pipeline, self-healing, and full-stack architecture
- Designed two-pass LLM pipeline: Stagehand browser agent explores target app and records ActionLog, converted to Gherkin feature files and Playwright step definitions
- Implemented self-healing mechanism: on test failure, LLM auto-patches selectors and timeouts and retries, improving generated test usability
- Full-stack architecture: React + Vite frontend, Fastify + Node.js backend, BullMQ + Redis job queue, Docker Compose, pnpm monorepo; generated output includes GitHub Actions workflow with zero LLM dependency at runtime

Education

Wenzao Ursuline University of Languages

Sep 2009 – Jun 2013

B.A. Digital Content Application and Management

Skills

Frontend	React / Next.js / TypeScript / Redux / GraphQL / Tailwind CSS / Vue 3 / Nuxt.js
Architecture & Infra	monorepo / CI/CD / GitHub Actions / GitLab Runner / AWS / Docker
AI Workflow	Claude Code / MCP / Playwright / Windsurf
Backend	Node.js / Rails / Koa
Mobile	React Native / Code Push / Fabric / TestFlight
Testing	Jest / Cucumber / RSpec